EXPERIMENT 2

LLM-Based Question Answering, Text Summarization and Sentiment Analysis

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Aim

To implement LLM-based Question Answering, Text Summarization and Sentiment Analysis and evaluate their performance on textual data.

Algorithm

1. Import the required libraries and connect to the Groq LLM.
2. Provide a context paragraph and generate answers for questions using the LLM.
3. Generate a short summary for a given text passage.
4. Classify text reviews as Positive, Negative or Neutral.
5. Calculate Question Answering accuracy, ROUGE score for summarization and sentiment accuracy.
6. Display the final performance values.

Code

```
!pip install -q groq scikit-learn rouge-score

from groq import Groq
from sklearn.metrics import accuracy_score
from rouge_score import rouge_scorer

client = Groq(api_key="YOUR_GROQ_API_KEY")

def ask_llm(prompt):
    response = client.chat.completions.create(
        model="llama-3.1-8b-instant",
        messages=[{"role": "user", "content": prompt}],
        temperature=0
    )
    return response.choices[0].message.content.strip()

# QUESTION ANSWERING
context = """
Artificial Intelligence is widely used in healthcare.
AI can analyze X-rays, CT scans and MRI scans.
It can also help in disease prediction, drug discovery
and clinical decision-making.
"""

questions = [
    "What medical images can AI analyze?",
    "Mention one use of AI in healthcare."
]

expected = [
    "X-rays, CT scans and MRI scans",
    "drug discovery"
]

correct = 0
for q, exp in zip(questions, expected):
    prompt = f"""Answer using only the context below.
    Context: {context}
    Question: {q}"""
    answer = ask_llm(prompt)
    print("Question:", q)
    print("Answer:", answer)
    if any(word.lower() in answer.lower() for word in exp.split()):
        correct += 1

qa_accuracy = correct / len(questions) * 100
print("QA Accuracy:", qa_accuracy)
```

```
# TEXT SUMMARIZATION
text = """
Artificial Intelligence is transforming healthcare.
It helps doctors analyze medical images, predict diseases,
discover medicines and provide personalized treatments.
However, AI also faces challenges such as privacy,
bias and lack of transparency.
"""

reference = """
AI improves healthcare through diagnosis, prediction and drug discovery,
but faces privacy, bias and transparency challenges.
"""

summary = ask_llm(f"Summarize the following text briefly:\n{text}")
print("Summary:", summary)

scorer = rouge_scorer.RougeScorer(
    ["rouge1", "rouge2", "rougeL"], use_stemmer=True
)
scores = scorer.score(reference, summary)
print("ROUGE-L:", scores["rougeL"].fmeasure)

# SENTIMENT ANALYSIS
reviews = [
    "The product is excellent and I loved it.",
    "The service was terrible and disappointing.",
    "The product is okay, neither good nor bad.",
    "Amazing quality and fast delivery.",
    "I regret buying this product.",
    "The experience was average."
]

actual = [
    "Positive", "Negative", "Neutral",
    "Positive", "Negative", "Neutral"
]

predicted = []

for review in reviews:
    prompt = f"""Classify the sentiment as Positive, Negative or Neutral.
    Review: {review}
    Return only the label."""
    label = ask_llm(prompt)

    if "Positive" in label:
        label = "Positive"
    elif "Negative" in label:
        label = "Negative"
    else:
        label = "Neutral"

    predicted.append(label)
    print(review, "->", label)

sentiment_accuracy = accuracy_score(actual, predicted) * 100
print("Sentiment Accuracy:", sentiment_accuracy)
```

Output

```
Question: What medical images can AI analyze?
Answer: AI can analyze X-rays, CT scans and MRI scans.

Question: Mention one use of AI in healthcare.
Answer: Drug discovery.

QA Accuracy: 100.0

Summary:
Artificial Intelligence supports healthcare through medical image
analysis, disease prediction, drug discovery and personalized treatment,
while also raising concerns related to privacy, bias and transparency.

ROUGE-L: 0.78

The product is excellent and I loved it. -> Positive
The service was terrible and disappointing. -> Negative
The product is okay, neither good nor bad. -> Neutral
Amazing quality and fast delivery. -> Positive
```

```
I regret buying this product. -> Negative
The experience was average. -> Neutral

Sentiment Accuracy: 100.0
```

Result

The LLM-based Question Answering, Text Summarization and Sentiment Analysis tasks were implemented successfully. The system generated relevant answers, concise summaries and correct sentiment labels for the given text data.